

IAS PRELIMS/MAINS



SCIENCE & TECHNOLOGY



A COMPREHENSIVE COVERAGE

MODULE - 38

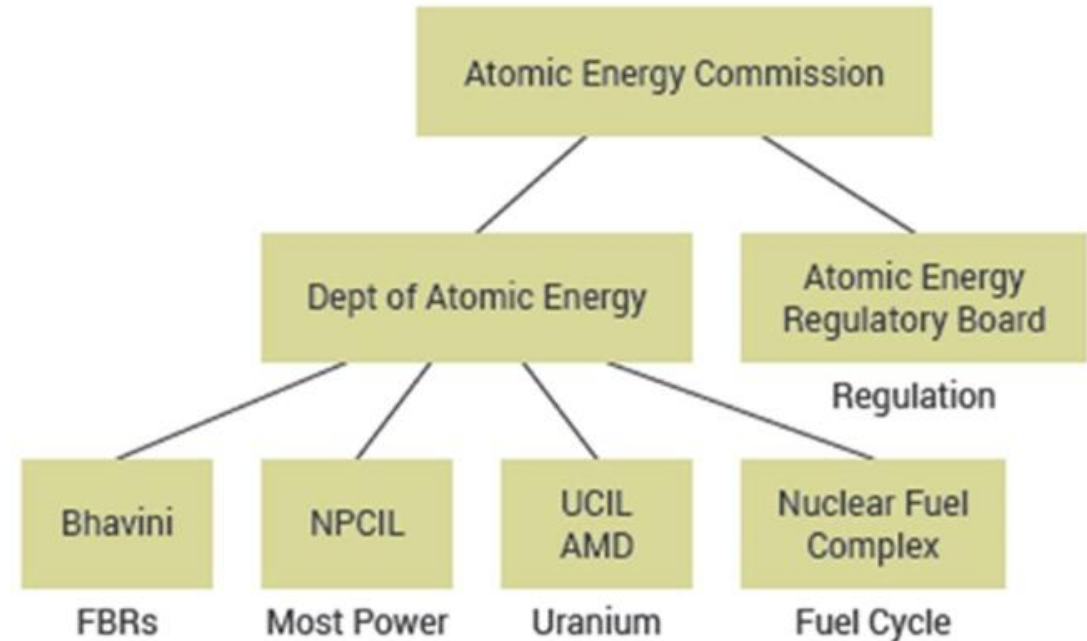
NUCLEAR TECHNOLOGY - 4

1

ATOMIC ENERGY COMMISSION

- The AEC was established in August 1948 within the Department of Scientific Research (established in June 1948). The Department of Atomic Energy (DAE) came into existence in 1954, thereafter it was transferred within the AEC.
- The Secretary to GOI in the DAE is the ex-officio Chairman of the AEC. The other members of the AEC are appointed on the recommendation of the Chairman of AEC.
- **The functions of the Atomic Energy Commission are:**
 - to organize research in atomic science in the country
 - to train atomic scientists in the country
 - to promote nuclear research in commission's own laboratories in India
 - to undertake prospecting of atomic minerals in India and to extract such minerals for use on industrial scale.

India Structure



Source: World Nuclear Association

NUCLEAR GOVERNANCE STRUCTURE IN INDIA

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DEPARTMENT OF ATOMIC ENERGY

Since its establishment, DAE has been engaged in the development of nuclear power technology, applications of nuclear technologies in the fields of agriculture, medicine, industry and research. DAE comprises five research centres, three industrial organisations, five public sector undertakings and three educational and research centers.

- The important programmes of the DAE are
 - ✓ Enhancing the share of nuclear power by deployment of indigenous and other proven technologies, and to develop fast breeder reactors, as well as thorium-based reactors.
 - ✓ Building and operating of research reactors for the production of radioisotopes and developing and deploying nuclear technology applications in the fields of medicine, agriculture, industry and basic research.



NUCLEAR GOVERNANCE STRUCTURE IN INDIA

2

DEPARTMENT OF ATOMIC ENERGY

- ✓ Developing advanced technologies such as **accelerators**, lasers, **supercomputers**, robotics, areas related to fusion research, strategic materials and instrumentation, and encouraging the transfer of technology to industry.
- ✓ **Carrying out** and supporting basic **research in nuclear energy** and related areas of science; interaction with universities and academic institutions; support to research and development projects and international cooperation in areas of research and contribution to national security.



ORGANIZATIONS UNDER DAE

RESEARCH & DEVELOPMENT

- The organisation that are there under DAE are
 - ✓ **Bhabha Atomic Research Centre (BARC), Mumbai.**
 - ✓ Atomic Minerals Directorate for Exploration and Research (AMD), Hyderabad.
 - ✓ **Indira Gandhi Centre for Atomic Research (IGCAR), Kalpakkam.**
 - ✓ Raja Ramanna Centre for Advanced Technology (RRCAT), Indore.
 - ✓ Variable Energy Cyclotron Centre (VECC), Kolkata.
 - ✓ Global Centre for Nuclear Energy Partnership

INDUSTRIAL ORGANISATIONS

- Heavy Water Board (HWB), Mumbai
- **Nuclear Fuel Complex (NFC), Hyderabad**
- Board of Radiation & Isotope Technology (BRIT), Mumbai



ORGANIZATIONS UNDER DAE

PUBLIC SECTOR UNDERTAKINGS

- **Electronics Corporation of India (ECIL), Hyderabad.**
- Indian Rare Earths Limited (IREL), Mumbai.
- Uranium Corporation of India, Singhbhum.
- **Nuclear Power Corporation of India (NPCIL), Mumbai.**
- Bharatiya Nabhkiya Vidyut Nigam Limited (BHAVINI), Kalpakkam.



EDUCATIONAL CENTERS

- Homi Bhabha National Institute, Mumbai
- **Tata Institute of Fundamental Research, Mumbai**
- Tata Institute of Fundamental Research, Hyderabad



NUCLEAR GOVERNANCE STRUCTURE IN INDIA

3

ATOMIC ENERGY REGULATORY BOARD

- AERB is responsible for regulating and monitoring the safety aspects of DAE's projects and programmes. DAE provides the technical, financial and staff support to AERB.
- The AERB is a regulatory body, it reports to the Secretary, DAE.
- The AERB reviews the safety and security of the country's nuclear power plants and nuclear development and research centers. The power of AERB is derived from the Atomic Energy Act, 1962.
- The mission of AERB is to ensure that the nuclear radiation does not cause any risk to human health and environment.
- The performance of Indian nuclear power reactors in the aspect of safety has been excellent. The release of radioactive particles from the reactors into the environment has been well within the safety limits prescribed by the AERB.



NUCLEAR GOVERNANCE STRUCTURE IN INDIA

4

NUCLEAR POWER CORPORATION OF INDIA LTD

- NPCIL is a public sector enterprise under the administrative control of the DAE. **NPCIL is responsible for the design, construction and operation of nuclear power plants.**
- The NPCIL was created in 1987, with the objective of undertaking the design, construction, operation and maintenance of the atomic power stations for generation of electricity.
- **NPCIL is presently operating 22 nuclear power plants with an installed capacity 5780 MW.**



NUCLEAR GOVERNANCE STRUCTURE IN INDIA

GLOBAL CENTRE FOR NUCLEAR ENERGY PARTNERSHIP

It will be the DAE's 6th R&D facility. It is being built in Haryana for strengthening India's collaboration internationally.

URANIUM CORPORATION OF INDIA LTD

UCIL is under the DAE, it was established in 1967 for uranium mining and uranium processing. **UCIL operates all the uranium mines in India.**



DAE'S TARGET

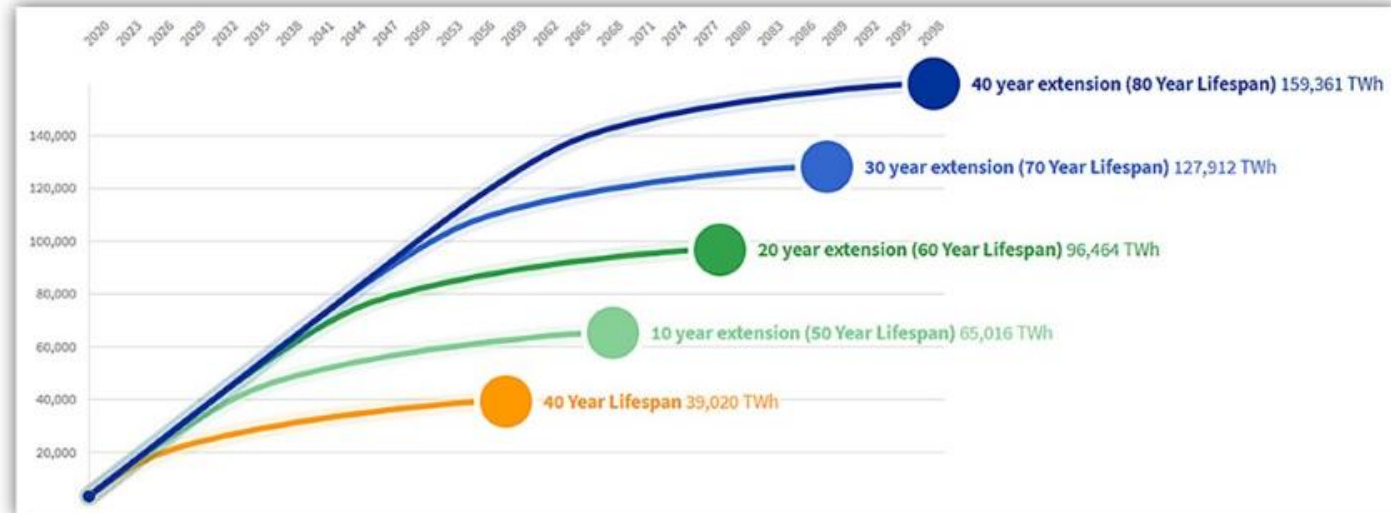
- The target set by DAE is, by 2050, 33% of India's total electricity requirement should be met by nuclear energy. This amounts to about 275000 MW.
- This target seems unachievable because of the following concerns—
 - ✓ India's domestic Uranium reserves can only support about 10000 MW of power. So, our future nuclear potential depends upon the development of the Third Stage of the Nuclear Program.

Otherwise, there will be overdependence on imported Uranium as it is the case with oil currently.
 - ✓ The nuclear reactors consume huge amount of water. So, most of the upcoming plants will be set up near the coasts. This will put pressure on the fragile coastal ecosystems of India.



DAE'S TARGET

- ✓ New safety requirements post **Fukushima disaster** has pushed per MW cost of nuclear reactors significantly higher in comparison to thermal, solar and wind plants.
- ✓ Also, till now only **21 plants are in operation**. And there are **long gestational periods for nuclear plants**.
- ✓ Some experts argue that the total cost of nuclear lifecycle – mining of uranium, transportation and storage, running costs of nuclear plants, possible disasters and handling of the waste generated is significantly more than the economic value generated during the **lifetime of a plant which is generally 40-50 years**.



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